Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 29 Introduction to Beams

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 29 Introduction to Beams

1 CLASS 29 Introduction to Beams

Study focus: CLASS 29 Introduction to Beams. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

When you are in the basement of any multi-story buildings like schools, apartments, or even in Mars, you can see some part of the building will be projected below the roofing, just like this, like a horizontal member.

Key details

- You can see in this picture these horizontal members, those projections, what you can see in the basement, we call them as beams.
- These beams play very important role in all the structures.
- Then how to define these beams?
- We are going to define beam as a structural member which offer resistance to bending at the rotation due to the applied loads.
- And these beams are usually long prismatic bars and the loads are applied normal to the axis of the bar.
- The first one is roller support.
- This is the roller support here.
- You can see the beam is resting on this roller support.
- The reason is if you think of applying a.
- Vertical load here, we can't move the beam along y direction because the roller will restrict the movement of this beam along the y direction.
- Whereas if you think you are going to apply a horizontal force on this beam like this, definitely it is going to move along the horizontal direction.
- Because there won't be any restriction from the support onto the movement of the beam along X direction.

- If you think of applying a couple at this free end, it is free to rotate about this roller support.
- That's why.
- Remember whenever we have a roller support, roller will offer resistance for the motion only along y direction, that's why we will be having only one reaction normal to the surface of the contact.
- This is about roller support.
- Next one is fixed support.
- In fixed support, the beam will be like this.
- It is represented in this way.
- Beam along y direction.
- At the same time, if you think of applying a horizontal force on this beam, again the.
- Support will offer restriction for the motion of the beam along X direction.
- That's why it will be having two reactions.
- One will be along X direction, one more will be along Y direction.
- In the same way, if you think you are going to apply a couple here at the free end R due to the application of the force.
- Do you think the beam will start to rotate like this about the axis of the beam?
- No, it won't, because.
- This will have a resisting couple.
- Three reactions to our forces, one along X, one more along Y, and one more will be the resisting couple like this.
- Next one is spin arranged support.
- This support will offer resistance for the motion of the beam in both along x direction as well as along the y direction.
- That's why two reactions, one along x direction and one more along y direction.
- This slide is very important to solve the numericals.
- Next is different types of beams.
- We are going to broadly classify the beams as statically determinate beams and statically indeterminate beams.
- Statically determinate beams.
- We can find the support reactions using the conditions of equilibrium.
- And in one more way, if the beam is having just sufficient number of support reactions which are enough for the beam to be in equilibrium.
- That is also called as statically determinate beam.
- If the beam is having more.
- Number of support reactions that necessary for the stable equilibrium configuration that is called as statically indeterminate beam.
- The first one is simple beam.
- If one end of the beam is on a pin connection or a hinged connection, and if one end is an roller, that we call it as a simple beam.
- How many number of support reactions here?

Example or application from the lecture

- We have different before moving on to the different types of beams, let's know about different types of supports.
- Ah when we draw the free body diagram in the numericals, if there is a roller support, usually we consider there will be only one reaction from the roller support.
- Let's talk about statically determinate beams.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then at B it's roller support, so one reaction that is RB reaction at B.

Key details

- We can find all the three reactions using three conditions of equilibrium.
- That's why the name statically determinate B.
- This is the first one simple beam.
- The second one is cantilever beam.
- If one end of the beam is fixed like this and the other end is free, it is not resting on any of the support.
- It is called as a cantilever beam.
- Since it is a fixed support, there'll be two forces, one along its direction,.
- One more in a Y direction and one more is the resting momentum.
- And if the answer is positive, our assumption is wrong.
- The last one is combination.
- This is also called as overhanging beam because this part of the beam is extended beyond the support, hence it is called as overhanging beam.
- It is pined connection.
- Then it's a roller, only one reaction along Y direction.
- These three are the statically determinate beams.
- Next, we have statically indeterminate beams.
- Remember, in this we can't find all the support reactions using the three conditions of equilibrium.
- The first one is continuous beam.
- If the beam is having more than three support reactions, three are more than three support reactions, it is called as continuous beam.
- It is a pinned connection, so two reactions, one along X and one more along Y, AX, AY.
- Next at B it's a roller support, so only one reaction.
- At C again it's a roller support, so one more reaction.
- Total we have four reactions Ax, AY, RC, and RB.
- Hence it is statically indeterminate B.
- The next one is fixed B.
- If both ends of the beam are fixed, it is called as fixed beam.
- See here it is fixed here and here also it is fixed support.
- Hence it is called as fixed beam.

- How many number of support reactions?
- Fixed support will offer.
- Three reactions, two forces and one more is moment, and here also two reactions and one more is moment.
- As I said before, the direction depends upon us how we are and we can assume the direction of the forces as well as the moment.
- Next one is end supported cantilever.
- This is also called as propped cantilever.
- Nothing but if one end of the beam is fixed, the other end is resting on a roller, it is called as end supported cantilever, a propped cantilever.
- Since it is a fixed support, three reactions and at B it's a roller support, so one reaction RB.
- Total we have four reactions, hence it is statically indeterminate B.
- This is about different types of beams.
- Next, the different types of loading on beams.
- The first one is point load.
- This is also called as concentrated load.
- As the name itself indicates, if the force acting on a beam is acting at a particular point.
- are the area over which the force is acting is very small compared to the dimension of the beam.
- We call that beam as a point load or concentrated load.
- Look at this figure there is a simple beam and.
- The load is acting at a particular point here, hence it is called as a point load or concentrated load.
- When we solve the numericals, we want all the loads to be point loads for our calculation purpose.

Example or application from the lecture

- Let's count the number of support reactions.
- You can consider it as either clockwise or counterclockwise, no need to worry about the direction.
- If you consider clockwise.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

And this point load can be either inclined, vertical, or horizontal.

Key details

- Few times it can be horizontal.
- A few times it can be inclined.
- Whenever they give the inclined load, remember we know how to resolve the load along.
- Force along X and Y.
- Same concept here also we are going to apply and find the components of that load.

- Nothing but force along X and Y direction.
- The second one is UDL.
- This is also called as rectangular loading, uniformly distributed loading.
- We are going to represent the loading like this.
- The blue color is the beam, and this is the load that is acting on this beam.
- The intensity, this is the assume this is the starting point of the load and this is the end point of the load.
- From ear to ear, the intensity is uniform throughout, hence the name uniformly distributed load or rectangular loading, and here W stands for the height of this rectangle.
- And L stands for the base, nothing but the span of the UDL.
- Then, as I said before, we want all the loads to be point load or concentrated loads.
- Then how to convert this UDL into a point load.
- That point load should produce the same effect as that of this UDL on a beam.
- There should not be any net change on the.
- Then how to get that?
- For that to find the resultant of this UDL, there is a formula.
- Since it is in the form of rectangle, resultant will be equal to W into L nothing but base into height of the rectangle.
- Once you find the resultant, this is the.
- Point load R is equals to W into L.
- Where exactly this point load will be acting because UDL is acting over a span of L meters.
- Then wave wheels are acting.
- Remember, since it's the rectangle from the concept of centroid, it will be exactly at the midpoint L by 2 from here.
- L by 2 from here.
- Here this R which is acting at a distance of L by 2 will produce the same effect as that of the UDL acting over a span of L meters.
- That's what the meaning.
- U VL that is uniformly varying load.
- This is also called as triangular load.
- Look at this figure, there is a beam.
- This is a beam and there is a UVL.
- The intensity of the loading is varying from 0 to W over a span of L meters.
- And this W will be either in kilonewton per meter or Newton per millimeter.
- Next, as I said before, we want all the loads to be point or concentrated loads.
- Then how to find the resultant of this load?
- Remember, since it is in the form of triangle, we are going to use the formula of.
- Triangle half into base into height.
- The resultant of this will be half into base into height, that is W into L by 2.
- You will get the magnitude of the resultant of this loading.
- Next, where exactly that will be acting?
- From this side, it will be 2 times that of L by 3, and from here L by 3.
- That you should take it.
- If you are measuring from this side, it will be L by 3 from here to L by 3.

- Fourth one is combined loading.
- It's also called as trapezoidal loading.
- It's a combination of UDL and UVL, hence it is called as combined loading.
- Look at the figure here.
- There is a beam and.
- Ah there is a loading.
- At one hand the height of the loading is W1, at the other end it is W2, and the span over which the load is acting is L.
- We know the formula for rectangular loading and triangular loading.
- Just assume we are going to draw a horizontal line like this.

Example or application from the lecture

- And intensity of this loading can be either given in the form of kilonewton per meter or newton per millimeter, and L can be either in millimeter or meter, and L is the span of the UDL.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

For the time being, forget about the triangle.

Key details

- How to find the resultant of the rectangle?
- Base into height.
- What is height here?
- This is W1, so this is the height.
- Then what is the base?
- L so R1 stands for R1.
- One is nothing but we are going to assume the first loading is a rectangle.
- R1 is equal to W1 into L.
- We got the magnitude of the resultant of the rectangle.
- Next thing is where exactly this R1 will be acting, we know from the concept of centroid from ear to ear, L by 2.
- For time being forget about the rectangle.
- What is base here?
- It is the same.
- Then what about the height of this triangle?
- W2 is not the height because W2 is the height from Ear to B.
- Total height is W2.
- Then how to find the height of the triangle that will be equal to total W2 minus this one.
- W1 that will be the height.
- Triangle R2 is equals to half into W2 minus W1 into L.
- Next we should know the point of application or the action of this.

- The last one is general loading.
- In general loading, the intensity of the loading will be varying.
- And in the syllabus we have the numericals on point load, UDL, UVL, and combined loading.
- That's it for today's class.
- Try to solve the questions from the question bank and go through the assignment questions.
- If you come across any queries, you can post it in the PESU forum.

Example or application from the lecture

- Just consider the rectangle.
- If you consider triangle, to find the resultant of the triangle, we know the formula half into base into height.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 29 Introduction to Beams transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 29 Introduction to Beams transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

C Revision and retrieval practice